Ionuț-Vlad MODORANU

Location: Vienna, Austria
E-mails: **modoranu.ionut.vlad@hotmail.com** (personal), **ionut-vlad.modoranu@ista.ac.at** (institutional)
Website: **ionutmodo.github.io**
Google Scholar: **https://scholar.google.com/citations?user=N56bz4gAAAAJ&hl=en**
LinkedIn: **https://www.linkedin.com/in/ionut-vlad-modoranu**

MAIN PUBLICATIONS

- **Ionuț-Vlad Modoranu**, Mher Safaryan, Dan Alistarh
  MatryoshkaLoRA: Learning Accurate Hierarchical Low-Rank Representations for LLM Fine-Tuning
  Arxiv

- **Ionuț-Vlad Modoranu**, Philip Zmushko, Erik Schultheis, Mher Safaryan, Dan Alistarh
  DASH: Faster Shampoo via Batched Block Preconditioning and Efficient Inverse-Root Solvers
  Poster @ ICML 2026

- **Ionuț-Vlad Modoranu**, Mher Safaryan, Erik Schultheis, Max Ryabinin, Artem Chumachenko, Dan Alistarh
  Trion: FFT-based Dynamic Subspace Selection for Low-Rank Adaptive Optimization of Large Language Models
  Poster @ ICLR 2026

- **Ionuț-Vlad Modoranu**, M. Safaryan, G.. Malinovsky, E. Kurtic, T. Robert, P. Richtarik, Dan Alistarh
  MicroAdam: Accurate Adaptive Optimization with Low Space Overhead and Provable Convergence
  Poster @ NeurIPS 2024

- **Ionuț-Vlad Modoranu**, A. Kalinov, E. Kurtic, E. Frantar, Dan Alistarh
  Error Feedback Can Accurately Compress Preconditioners
  Poster @ ICML 2024

ACTIVELY MAINTAINED GITHUB PROJECTS

ISTA-DASLab-Optimizers

Managing a GitHub repository containing all stochastic optimizers for Deep Learning developed in our research group at ISTA. The goal is to provide an official PyTorch implementation and the associated CUDA kernels for each optimization method, allowing easy experimentation and bug fixing. It is installable via **pip install ista-daslab-optimizers**.

GridSearcher

Developed a Python library to speed up the hyper-parameter tuning process. The user has to specify the parameter grids and GridSearcher launches jobs in parallel over the cartesian product of parameter grids and uses a basic GPU management approach. It is designed for machines where users have direct ssh access. It is installable via **pip install gridsearcher**.

STUDIES

September 2021 – present
Ph.D. student at Institute of Science and Technology Austria (ISTA)
Deep Algorithms and Systems Group, supervised by Dan Alistarh
Research topic: Efficient Optimization for Deep Learning (low-memory optimizers, advanced numerical linear algebra techniques, low-rank optimizer states)

October 2019 – February 2020
Erasmus+ mobility (Master's level)
Faculty of Computer Science, Mathematics and Applied Mathematics
University Grenoble Alpes, Grenoble, France

October 2018 – July 2020
Master's of Advanced Studies in Computer Science (Research)
Faculty of Computer Science
University "Alexandru Ioan Cuza" from Iași, Romania

- 1st year grade: 9.75 / 10
- 2nd year grade: 9.10 / 10

Master's thesis:
A new adversarial attack that slows down decision making process in multi-exit deep neural networks is the result of the internship project from summer of 2019 started at University of Maryland and continued in March-July 2020. As a result, this work turned into a research paper published at **ICLR 2021**.

October 2014 – July 2018
Bachelor's Degree in Computer Engineering
Faculty of Automatic Control and Computer Engineering
Technical University "Gheorghe Asachi" Iași, Romania

- 1st year grade: 9.81 / 10
- 2nd year grade: 9.68 / 10
- 3rd year grade: 9.91 / 10
- 4th year grade: 9.76 / 10

Bachelor's thesis:
Traffic signs and road lanes detection and classification for an autonomous driving car at small scale.
Won 2nd place in the "Electromobility" student contest 2018 with a team of 4 members. Designed and implemented vision and steering functionalities of the car: used the Haar Cascade classifier to detect traffic signs and a Convolutional Neural Network (adapted from NVidia's paper) to obtain steering decisions. Two short videos with our self-driving car:

- on the track it was trained on: 1) https://youtu.be/6GwoaJXw8vQ, 2) https://youtu.be/RCGOVQTgiKQ
- on a new, previously unseen track: https://youtu.be/kIdHWN1AKY8?t=242

WORK EXPERIENCE

June 2025 - September 2025: Research Scientist Intern at Together AI
Developed an efficient, low-rank version of the Muon optimizer called Trion, based on a dynamic low-rank projection matrix that is much cheaper than SVD, accepted at ICLR 2026.

December 2020 – August 2021: Research Scientist at Amazon
Worked on Anomaly Detection from large databases. The project goal was to detect anomalous actions from a chain of discrete actions for each individual user of different services.

July 2019 – September 2019: Research Intern at University of Maryland (UMIACS), College Park, USA
Developed the initial version of DeepSloth, an adversarial attack designed to bypass the early-exits of a Convolutional Neural Network, published as a spotlight paper at ICLR 2021.

July 2018 – October 2018: Research Scientist Intern at Amazon
Part of the Cloud Tune team which was responsible for the AWS Auto Scaling Service. Trained DeepAR models (Recurrent Neural Network based on LSTM) for time series to obtain more accurate predictions used for the auto scaling service in AWS.

July 2017 – October 2017: Machine Learning Engineer Intern at Amazon
Part of Amazon Robotics team. Developed a model to predict the time to failure of robots inside fulfillment centers and classifying hardware/software errors that may occur during robots' operation, based on their previously logged information.